Interrupting Autopilot: Evaluating Sensory Interventions for Momentary Self-Regulation

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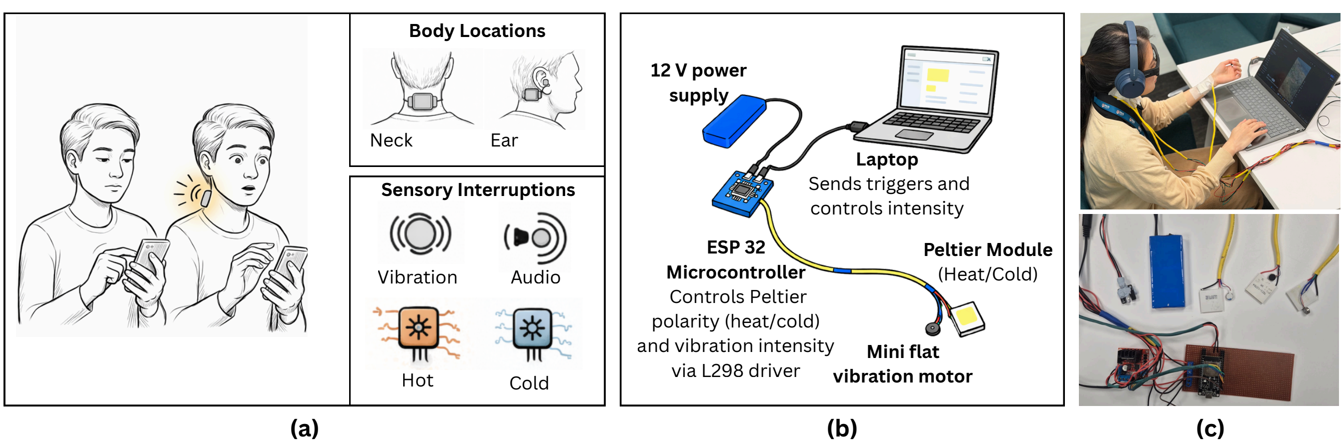

Figure 1: Overview of the sensory interruption system. (a) Sensory interruption modalities (vibrotactile, heat, cold, audio) and body locations (behind the neck and behind the ear). (b) Hardware pipeline showing laptop-triggered control of vibrotactile and thermal cues via a microcontroller. (c) Prototype deployment during a short-form video browsing task.

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Abstract

People have different temptations and habitual behaviors they wish to change, often imagining a better version of themselves. However, most existing behavior-change interventions rely on conscious self-regulation. Existing tools require users to notice a prompt, interpret it, and deliberately override an impulse, which is cognitively demanding and prone to failure when behavior becomes automatic. In this paper, we explore sensory interruptions as a way to bridge perception, cognition, and behavior. These interruptions aim to momentarily disrupt autopilot behavior and enable users to regain control. We evaluated four sensory interruptions: vibrotactile, heat, cold, and audio, with 11 participants. Our findings reveal clear subjective trade-offs: vibrotactile was often perceived as habituated, while heat was effective but generally disliked due to its punitive feel. Nevertheless, most interruptions produced a brief micro-pause that prompted participants to reconsider their behavior, highlighting the potential of physical modulation as a complementary approach to self-regulation.

CCS Concepts

• **Human-centered computing** → **Interaction devices**; **Interaction techniques**.

Keywords

Subconscious Interrupt, Context Aware Nudging, Habit Forming, Assistive Augmentation

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1 Introduction

People often desire greater control over everyday habits and seek to reduce behaviors they consider unhelpful. A common example is mindless scrolling through short-form video content on a digital device, often referred to as doomscrolling, in which continuous reading and scrolling occur with little intentional control. Such behaviors typically occur automatically and with minimal conscious awareness. Consequently, self-regulation often relies on willpower and conscious effort, which are known to be fragile and inconsistent when confronting automatic, repetitive behaviors [2, 8, 22]. This gap between intention and action remains a central challenge in behavior change research [21, 24].

Many existing digital self-regulation tools rely on explicit nudges, such as smartphone notifications or smartwatch vibrations [21, 24]. These approaches share a common assumption that users will notice the cue and deliberately choose to act. However, this assumption often fails when behavior is already automatic. Such nudges are frequently ignored or quickly habituated to, particularly when users are already engaged in what is often described as "autopilot" behavior [3, 4]. This reveals a fundamental mismatch: interventions that require deliberate attention struggle to interrupt behaviors that bypass conscious awareness [2, 9, 16].

Therefore, *how can interventions support self-regulation when behavior is driven primarily by automatic processes rather than conscious decision-making?* We propose shifting the point of intervention from cognition to perception. Instead of relying on deliberate attention, we explore whether habitual behavior can be interrupted at a bodily level through brief sensory stimulation. We refer to this approach as *sensory interruption*.

This approach is informed by two key insights from prior research. First, situational interventions tend to be more effective than cognitive strategies for addressing automatic behavior. Interruptions embedded in the environment can reduce mindless behavior by creating a brief pause, without requiring explicit instruction or reflection [2, 9]. Second, the timing and mechanism of intervention are critical. Interventions that explicitly draw attention to decision-making can sometimes backfire, increasing rather than reducing undesirable behavior [21]. From a self-control perspective, attempting to actively inhibit an ongoing impulse is difficult and often fails once automatic behavior has begun. Proactive strategies that redirect attention *before* full engagement are therefore more reliable [15].

Bodily sensations offer a promising mechanism for such early intervention. By directly engaging perception, they may interrupt habitual engagement *before* conscious deliberation is required. A brief sensation can produce a perceptual shift that naturally redirects attention away from the habitual activity, creating an opportunity for intentional choice.

To investigate this sensory interruption approach, we conducted an exploratory study examining four types of sensory interruptions: haptic, cold, heat, and audio. With the exception of audio, these sensations were delivered at two body locations behind the neck and behind the ear. These locations were selected for three reasons: (1) their sensitivity to subtle stimulation [17, 20], (2) their minimal interference with ongoing actions and task performance, and (3) their feasibility for integration with emerging wearable technologies such as smart glasses, which make detecting moments of habitual engagement increasingly feasible. Our study examines participants' observable reactions and subjective experiences, alongside objective measures such as reaction time and response speed following interruption. In this paper, we focus our analysis on subjective reports and qualitative observations, using objective measures as supporting signals that we leave for deeper analysis in future work.

The goal of this pilot study is not to claim long-term behavior change, but to provide early empirical insights into how different sensory interruptions are perceived and how they influence moment-to-moment disengagement from automatic behavior. By characterizing both qualitative and quantitative responses, this work lays the groundwork for future research on bodily-sensation-based interventions that support intentional self-regulation.

2 Related Work

Doomscrolling as a Habit to Interrupt. Our motivation is to understand how habitual digital behaviors can be interrupted and redirected toward healthier alternatives. We focus on *doomscrolling*, which is the prolonged and automatic consumption of short-form social media content, this behaviour has become increasingly common and difficult to regulate. Prior work characterizes such behavior as highly immersive, with increased visual engagement and reduced disengagement, suggesting strong attentional capture [6, 12]. To study this phenomenon in a controlled setting, we asked participants to freely scroll through YouTube Shorts, using doomscrolling as a representative habitual behavior.

Bodily Sensations as Interruption Mechanisms. Most existing approaches to reducing problematic digital habits rely on explicit notifications, screen-time tracking, or interface-level interventions, which require conscious user attention and deliberate self-control [14, 18, 19, 21]. These approaches are often less effective once behavior becomes automatic.

An alternative approach is to intervene through bodily sensations. Vibrotactile feedback has been widely explored for behavior change, including posture correction and reducing sedentary behavior, but shares similarities with conventional notifications and may be prone to habituation [5, 7, 10, 23]. Thermal feedback has been studied primarily in safety and warning contexts, yet remains underexplored as a mechanism for interrupting habitual digital behavior. Cognitive research suggests that bodily sensations can capture attention rapidly and preconsciously, creating brief opportunities to disrupt automatic behavior before effortful self-control is required [2, 9]. We refer to this brief disruption as a *micro-pause.* In this work, we compare vibrotactile, thermal, and audio cues to examine how different interruption modalities balance effectiveness, comfort, and long-term usability.

3 Study Design and System Overview

We designed the study as a within-subject mixed-methods exploration combining subjective reports, qualitative observations, and objective sensing. The primary goal was to understand how different sensory interruptions are perceived and how they influence moment-to-moment disengagement from habitual behaviors.

Sensory Interruption System. To deliver sensory interruptions, we developed a custom actuation setup consisting of lightweight wearable modules capable of producing thermal (heat and cold) and vibrotactile stimulation, as well as audio cues delivered through earphones. Thermal stimulation was achieved using Peltier elements, where reversing the direction of current produced heating or cooling, while vibrotactile feedback was provided through a compact flat vibration motor. All components were controlled by a central unit (laptop) responsible for triggering, synchronization, intensity control, and timestamping of actuation events. Figure 1 provides an overview of the study setup. The wearable modules were attached to different body locations using medical-grade adhesive patches.

Procedure. Participants were first screened to ensure that excessive short-form video consumption was a behavior they wished to regulate. They then browsed YouTube Shorts on a laptop, selecting content aligned with their personal interests to approximate natural infinite scrolling behavior.

The wearable modules were attached to three body locations: the wrist, behind the ear (mastoid region), and the back of the neck (C7 vertebral area) [17]. The wrist was used as a baseline location to calibrate individual tolerance levels for thermal stimulation before applying sensations to more sensitive areas. Ear and neck placements were counterbalanced across participants. Sensory modalities (vibrotactile, heat, and cold) were randomized within blocks to reduce ordering effects. All actuation events were logged to support post-hoc analysis. Sensory interruptions were delivered opportunistically during moments of high engagement (e.g., leaning closer to the screen or visibly reacting to content), rather than at fixed time intervals. Further details on the study protocol, hardware implementation, and safety procedures are provided in the supplementary materials.

All participants wore Aria glasses [11] during the session. This served two purposes: (1) to collect objective measures such as head movement and gaze for potential future analysis, and (2) to maintain the ecological validity of wearing an assistive or sensing wearable during everyday interaction. After each block of sensory interruptions, participants completed a short questionnaire consisting of three Likert-scale items (1–7), assessing noticeability, perceived effectiveness in interrupting behavior, and willingness to use the interruption in everyday life. Each participant experienced seven interruption conditions in total, covering combinations of modality (audio, vibrotactile, heat, cold) and body location (neck, ear). Following the completion of all conditions, we conducted a semi-structured interview to gather qualitative feedback. Participants were asked to reflect on their preferences across modalities and locations, compare bodily interruptions to familiar audio or wrist-based notifications, and discuss contexts in which they would or would not want such a wearable to intervene. These interviews, together with in-situ observations, informed the qualitative analysis reported in the following Section.

4 Preliminary Analysis and Discussion

We conducted the study with 11 participants (M=8, F=3), Six participants reported already using active strategies to regulate doomscrolling, most commonly uninstalling or deleting applications, or using third-party tools to track usage or impose time limits. Given the small sample size (N=11) and the exploratory nature of the study, we report descriptive statistics and visual trends rather than inferential statistics. The goal of this analysis is not to establish statistical significance, but to establish comparative patterns and trade-offs across modalities and body placements that can inform future design and hypothesis-driven work. We complement these quantitative trends with qualitative data to help contextualize and interpret participants' responses.

Quantitative Analysis of the Subjective Ratings. Figure 2 summarizes participants' subjective ratings across three dimensions: noticeability, perceived interruption (reconsideration), and everyday acceptability. Overall, the box plots show that most sensory interruptions were clearly noticed, with median ratings above the neutral midpoint across modalities and body locations. Haptic and cold interruptions generally received higher median ratings for everyday acceptability, while heat-based interruptions showed greater variability and lower acceptability despite being rated as effective for prompting reconsideration. Audio interruptions produced moderate ratings but exhibited overlap with familiar notification patterns. Further to this, Figure 3 presents ratings aggregated by cue modality. Consistent with the placement-specific results, heat-based cues were rated as effective for behavioral reconsideration but less suitable for everyday use, suggesting potential concerns around comfort or long-term adoption. An analysis across body placements (ear vs. neck) did not reveal substantial systematic differences.

Qualitative Feedback and Observations.

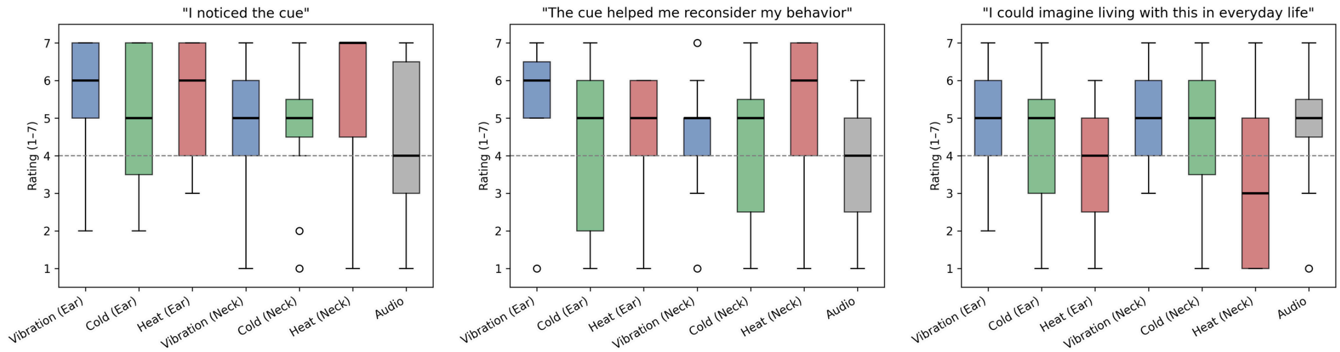

Figure 2: Participant ratings (median and interquartile range) for the everyday acceptability of each sensory interruption. Colors indicate interruption modality (vibrotactile, cold, heat, audio); ear and neck locations are indicated in the labels.

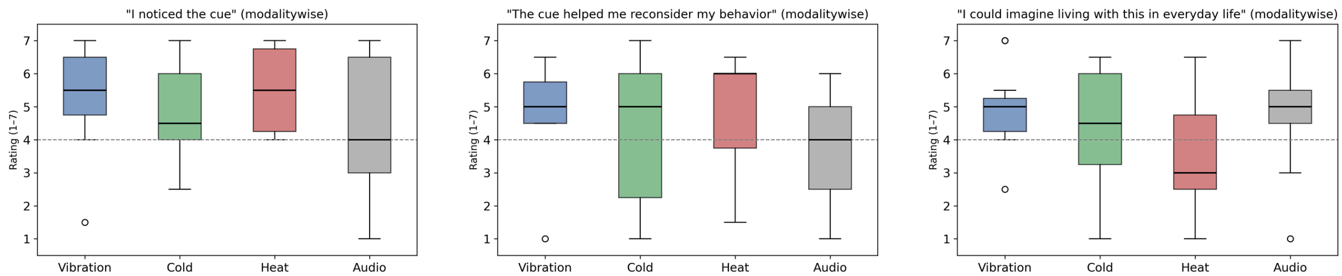

Figure 3: Participant ratings comparing cue modalities (averaged across neck and ear placements, where applicable) across three subjective evaluation measures.

Sensory interruptions reliably produce a brief "micro-pause," even during moments of high engagement. Across conditions, we observed a consistent behavioral response to sensory interruptions. Regardless of modality or placement, participants almost always shifted their gaze away from the screen immediately following an interruption. Interruptions were delivered opportunistically, timed to moments when participants appeared highly engaged in the content (e.g., leaning closer to the screen [1, 13] or laughing). In these moments, the interruption reliably produced a brief disengagement, suggesting the presence of a momentary "micro-pause" in otherwise continuous, habitual interaction. In Figure 4 we provide a qualitative illustration of participants' immediate reactions to different sensory interruption modalities and body locations, that foregrounds the subjective variability that complements our quantitative analysis.

Effectiveness and user preference are distinct dimensions that can diverge. Qualitative feedback revealed a consistent trade-off between interruptiveness and comfort across sensory interruptions. Heat-based interruptions were frequently described as highly effective at immediately stopping doomscrolling behavior, often evoking strong reactions such as discomfort or a sense of danger. Several participants framed heat as punitive, noting that it forced disengagement but would be undesirable for repeated everyday use (e.g., *"it feels like a punishment"* [P6] and *"I want to stop it immediately"* [P4]). Importantly, participants distinguished between effectiveness and preference as separate dimensions. For example, one participant noted, *"I think the most effective is heat, but I prefer vibration… heat makes me want to get rid of it"* [P4]. In contrast, cold and haptic sensations were generally perceived as more acceptable and less aversive, even when their interruptive effect was described as more gradual. This suggests that while strong aversive cues can reliably disrupt behavior, they may reduce long-term acceptability.

Habituation reduces the effectiveness of familiar notification-based cues over time. Audio and wrist-based vibrotactile were frequently compared to familiar notifications (e.g., alarms or smartwatch alerts) and described as easy to ignore due to habituation. This suggests that habituation may shape the long-term effectiveness of different interruption modalities: while some interruptions may become more tolerable and less disruptive over time, others risk becoming habits themselves and losing their interruptive power. These observations point to the need for further investigation into how interruptive cues perform over extended use.

Less cognitively mediated, bodily cues may interrupt behavior before conscious dismissal occurs. Several participants described heat and cold sensations as operating at a more subconscious level, prompting an immediate shift in attention without deliberate reasoning (e.g., *"there was no thinking and I looked away"* [P5]). Such less cognitively mediated cues may therefore offer an alternative pathway for disruption that is less susceptible to conscious dismissal. At the same time, participants highlighted the contextual

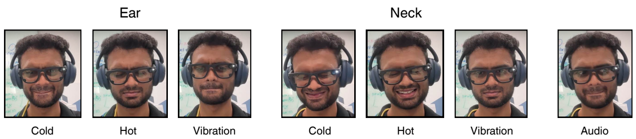

Figure 4: Representative participant facial reactions to different sensory interruption stimulated behind the ear and the neck

nature of these sensations. For example, one participant noted that preferences for heat or cold could depend on environmental conditions, suggesting that the effectiveness and acceptability of thermal interruptions may vary with situational context (e.g., weather or ambient temperature).

Preferences for body location varied widely across participants. The ear was often described as highly sensitive, whereas the neck was perceived as effective but strongly context-dependent (e.g., uncomfortable when lying down [P4, P7]). Across participants, situational appropriateness played a key role in acceptance, reinforcing the importance of designing sensory interruptions that are not only perceptually distinct but also adaptable to context, placement, and individual tolerance. Together, these findings highlight modality novelty, bodily placement, and habituation as critical considerations when designing such sensory interruptions to disrupt habitual engagement.

5 Limitations and Future Work

While the current study was sufficient to surface comparative patterns and user perceptions, it limits our ability to draw conclusions about long-term effectiveness, habituation, or sustained behavior change. Future studies should examine repeated use over extended periods to better understand how sensory interruptions evolve with familiarity and whether their impact persists or diminishes over time. Our findings suggest a trade-off between long-term acceptability and effectiveness. While vibrotactile-based interventions may be quickly habituated and ignored over time, heat-based cues despite their lower tolerance, may encourage users to actively change their behavior rather than adapt to the intervention, potentially making them more effective at breaking habitual behavior loops.

In addition, although we collected objective measures such as gaze, head movement, and response timing, the present analysis focused primarily on subjective experience and qualitative observations. Future work will integrate these objective signals to quantify micro-pausation, disengagement latency, and recovery dynamics following interruptions. Such analysis could enable more precise characterization of how different modalities influence attention at a temporal level.

Finally, participants' preferences varied widely across modalities and body locations, and were often shaped by situational context (e.g., posture, environment, ongoing activity). This highlights the need for adaptive and personalized interruption systems that account for individual sensitivity, context, and environmental conditions. Future systems could dynamically adjust modality, intensity, and placement to balance effectiveness and comfort. Together, these directions can inform the design of sensory interruptions that nudge users towards their ideal behavior.

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